Mechatronic Sensors Trainer

User Manual – Setup and Configuration

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For more information on the solutions Quanser offers,
please visit the web site at: http://www.quanser.com

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Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

FCC Notice This device complies with Part 15 of the FCC rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.
Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

Industry Canada Notice This Class A digital apparatus complies with CAN ICES-3 (A). Cet appareil numérique de la classe A est conforme à la norme NMB-3 (A) du Canada.

Waste Electrical and Electronic Equipment (WEEE)
This symbol indicates that waste products must be disposed of separately from municipal household waste, according to Directive 2012/19/EU of the European Parliament and the Council on waste electrical and electronic equipment (WEEE). All products at the end of their life cycle must be sent to a WEEE collection and recycling center. Proper WEEE disposal reduces the environmental impact and the risk to human health due to potentially hazardous substances used in such equipment. Your cooperation in proper WEEE disposal will contribute to the effective usage of natural resources.

CE Compliance CE

This product meets the essential requirements of applicable European Directives as follows:

- 2014/30/EU; Electromagnetic Compatibility Directive (EMC)
- 2014/53/EU; Radio Equipment Directive (RED)

Warning: This is a Class A product. In a domestic environment this product may cause radio interference, in which case the user may be required to take adequate measures.

Caution

Laser Safety: The STMicroelectronics VL53L8CH Time-of-Flight (ToF) sensor contains an integrated laser emitter and drive circuitry. The laser output is designed to meet Class 1 laser safety limits under all reasonably foreseeable conditions, including single faults, and is in compliance with the following standards:

- IEC 60825-1:2014
- 21 CFR 1040.10 and 1040.11, except for conformance with IEC 60825-1 Ed. 3 as described in Laser Notice No. 56, dated May 8, 2019.
- EN 60825-1:2014 including EN 60825-1:2014/A11:2021
- EN 50689:2021 (with applicable exceptions as specified by the manufacturer)

Do not power on the product if any external damage is observed. Do not increase the laser output power by any means. Do not use optics to focus the laser beam. Do not open, disassemble, or modify the sensor/module; doing so may cause emissions to exceed Class 1. Invisible laser radiation may be present when opened. Do not look directly into the emitting aperture or view it through optical instruments (e.g., magnifying glass, microscope).
Caution: Use of controls, adjustments, or procedures other than those specified by Quanser and the device manufacturer may result in hazardous radiation exposure.

Caution

FCC Guidelines for Human Exposure: This product contains an Acconeer A111 radar sensor module operating in the 57–64 GHz band. This equipment complies with FCC radiation exposure limits set forth for an uncontrolled environment. When using the radar sensor, this equipment should be installed and operated with minimum distance of 20 cm between the radiator and your body. This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter. The A111 sensor is not permitted for use on satellites or aircraft.

Caution

ESD Warning: The Mechatronic Sensors Trainer internal components are sensitive to electrostatic discharge. Before handling the Mechatronic Sensors Trainer, ensure that you have been properly grounded.

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A. Introduction

The Quanser Mechatronic Sensors Trainer is an integrated sensor experiment. It is designed to help teach fundamental sensor concepts and theories on an easy-to-use and intuitive platform.

The compact device includes a rich suite of distance, motion, environmental, light, force, and touch sensors, allowing students to not only learn the fundamental workings of these sensors, but also develop key skills in component selection and making design decisions for complex measurement and perception systems.

With the ability to provide real-time measurements, the Mechatronic Sensors Trainer can be integrated into larger systems or used to enhance the functioning capabilities of other Quanser systems. In addition, this device can also be used to measure and store data asynchronously for offline processing at a later time.

The unit is intended for lab and classroom use to support fundamentals labs, guided integration challenges, and open-ended projects.

Figure 1. Mechatronic Sensors Trainer

B. Hardware Components

The main components of the Sensors Trainer are listed in Table 1 and 2. These components are marked in Figures 2 and 3, which show the front, and side views of the Trainer.

The datasheet for each of the components is located in the same folder as this user manual.

i. Parts of the Device

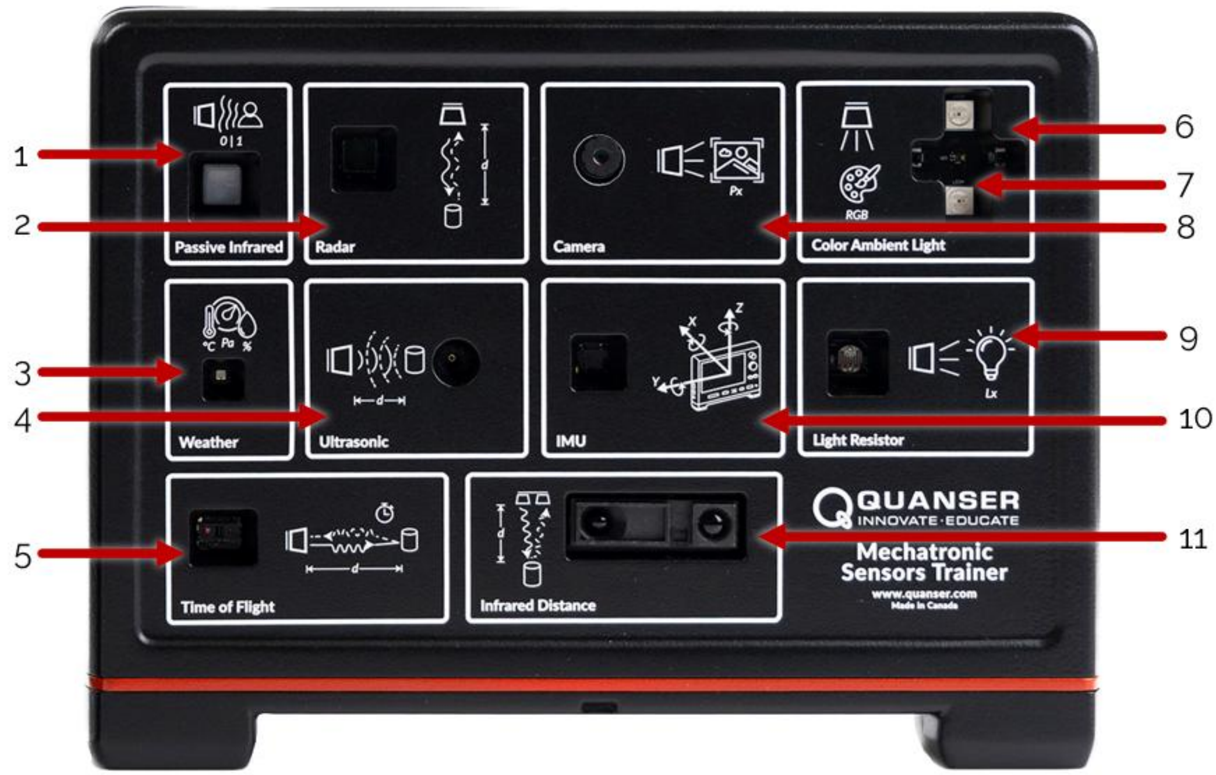


Figure 2. Front of Mechatronic Sensors Trainer

ID	Component	ID	Component
1	Passive Infrared Sensor and Cap	**6**	RGB and IR LEDs
2	Radar Sensor	**7**	Color Ambient Sensor
3	Weather Sensor	**8**	Camera
4	Ultrasonic Sensor with Cone	**9**	Light Dependent Resistor
5	Time of Flight Sensor	**10**	IMU Sensor
		11	Infrared Distance Sensor

Table 1. Components in the front of the device

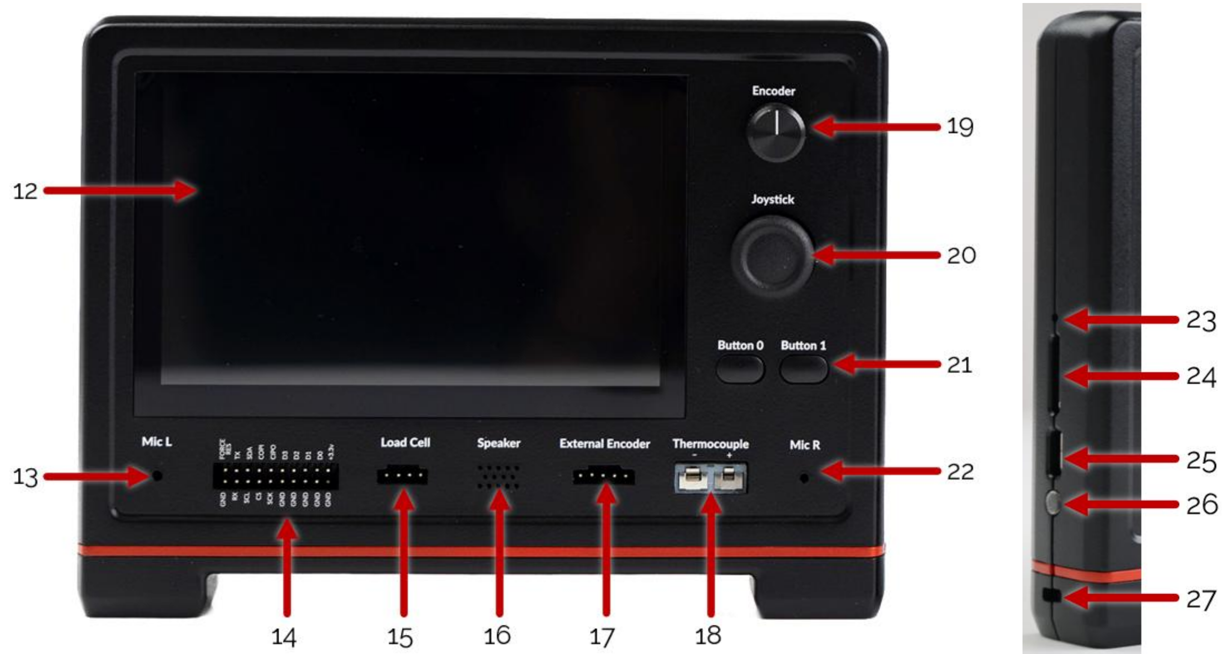


Figure 3. Back and Side of Mechatronic Sensors Trainer

ID	Component	ID	Component
12	LCD Touch Sensor	**20**	Joystick
13	Left Microphone	**21**	Push Buttons (0 and 1)
14	GPIO	**22**	Right Microphone
15	Load Cell Connector	**23**	Programming Pin
16	Speaker	**24**	SD Card Slot
17	External Encoder/PWM Connector	**25**	USB C Connector
18	K-Type Thermocouple Connector	**26**	Status LED
19	Encoder Knob	**27**	Standard (K type) Kensington Lock

Table 2. Components in the front and side of the device

ii. External Provided Parts

There are a few external provided parts as shown and labelled in the picture below. The Force Sensing Resistor (1) which gets connected to the leftmost pins of the GPIO. The Radar cone (2) fits into the Radar cutout on the sensor side of the device. The thermocouple (3) connected to the thermocouple connector under the LCD display; and the Loadcell which gets connected to the Load Cell pins under the LCD display. A Potentiometer is also provided which will be used in the potentiometer lab.

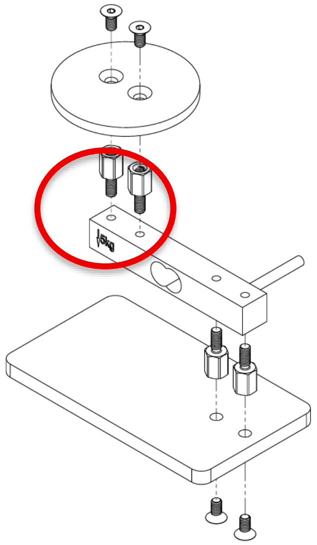
iii. Assembling the Load Cell into a Scale

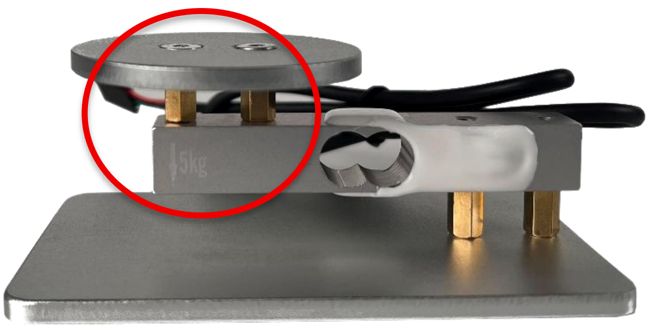
The provided pieces include parts to build the load cell into a scale. The pieces needed for this are:

- The load cell,
- the rectangular and circular metal plates,
- the 4 gold-colored standoffs,
- 4 flat, countersunk screws (M4)
- An M2.5 (2.5 mm) Allen Hex Drive (provided)

Please note when building the loadcell that the circular plate (measuring plate) will be placed above the side of the loadcell that shows the weight, so that the final scale has the arrow pointed down. Use the picture below as reference.

Use the images below as a reference on how to build the scale.



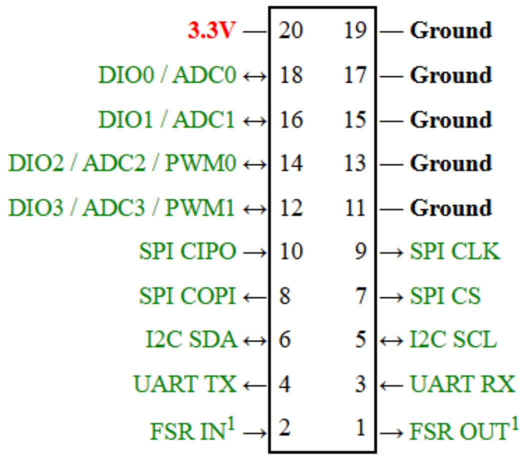


iv. Connectors and GPIO

The Quanser Mechatronic Sensors Trainer has a number of connectors for sensors or expansion I/O. These connectors and their pinouts are listed below.

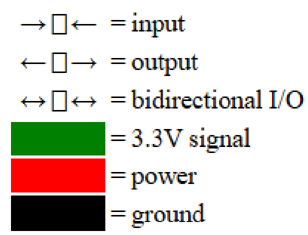
Note that all pins are 3.3V and are *not* 5V tolerant unless marked otherwise. The analog inputs only support 0 to +3.3V. *Exceeding these voltages may damage the board!*

User I/O Connector (GPIO)

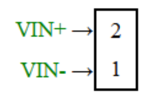


[1] Dedicated Force Sensing Resistor (FSR) pins

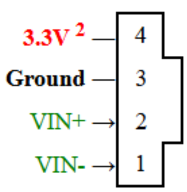
Legend



Thermocouple Connector

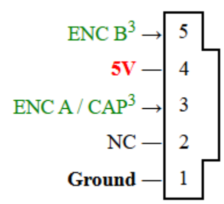


Load Cell Connector



[2] Dedicated Load Cell power supply. Do not use for anything else.

Encoder Connector



[3] 5V tolerant input pins

v. Dimensions

Dimension	Size
Length	18 cm
Width	3 cm
Height	13 cm

vi. Capabilities

The Quanser Mechatronic Sensors Trainer provides the following capabilities:

- Analog joystick with X and Y axes (16-bit resolution) and button
- Analog inputs (four optional single-ended channels 0.0V to 3.3V. 16-bit resolution)
- Camera (RGB, up to 2592x1944 resolution, OV5640 sensor)
- Color and ambient light sensor (including clear, R, G, B and infrared, VEML3328)
- CPU temperature
- Current sense on external +3.3V supply (INA226)
- Digital I/O (four channels)
- External encoder (with 1X, 2X and 4X quadrature)
- Force sensing resistor (16-bit resolution)
- IMU with 3-DOF gyroscope (up to 2000 °/s), 3-DOF accelerometer (up to ±16g), 3-DOF magnetometer (up to 5000 µT) and temperature (ICM-20948)
- Infrared distance sensor (16-bit resolution)
- Input capture measuring period and pulse width (one optional channel)
- Knob encoder (with 1X, 2X and 4X quadrature and A/B signals)
- LCD display with touch panel (800x480 resolution, 120 Hz touch panel up to 10 fingers)
- LEDs to provide illumination for the color sensor (two RGB and infrared LEDs)
- Light-dependent resistor (16-bit resolution)
- Load cell sensor (24-bit resolution, NAU7802)
- Passive infrared sensor (16-bit resolution, PL-Q873-02)
- Pushbuttons (two buttons with configurable polarity)
- PWM outputs (two optional channels up to 16-bit resolution and up to a 120 MHz PWM)
- Radar (includes distance, presence, power bins, envelope, IQ and sparse measurements, with up to 14455 points, A111-001-T&R)
- Speaker (96 kHz audio, MAX98357AETE+T)
- Stereo microphone (48 kHz 24-bit audio, ICS-41351)
- Thermocouple (internal & external temp, fault detection, MAX31855KASA+T)
- Time of flight sensor with up to 64 zones (includes distance, sigma, reflectance and targets detected per zone, VL53L8CH)
- UART for external serial devices (up to 15 Mbaud)
- Ultrasonic sensor (includes distance and up to 450 IQ data points, CH201-00ABR)
- Weather sensor (pressure, temperature with up to 20-bit resolution and humidity with 16-bit resolution, BME280)
- Game controller overrides for all point-of-view, buttons, joysticks and bumpers

vii. Mechatronic Sensors Trainer as Devices in a PC

When the Quanser Mechatronic Sensors Trainer is connected to a PC it will show up as multiple devices. In particular, it will appear in Device Manager in Windows as the following devices (or possible variations thereof):

- Audio inputs and outputs/Microphone (2-Quanser Mechatronic Sensors Trainer)
- Audio inputs and outputs/Speakers (Quanser Mechatronic Sensors Trainer)
- Cameras/Quanser Mechatronic Sensors Trainer
- Ports (COM & LPT)/USB Serial Device (COM*nn*)
- Universal Serial Bus devices/Quanser Mechatronic Sensors Trainer HIL
- Universal Serial Bus devices/Quanser Mechatronic Sensors Trainer I2C
- Universal Serial Bus devices/Quanser Mechatronic Sensors Trainer LCD
- Universal Serial Bus devices/Quanser Mechatronic Sensors Trainer SPI
- Xbox 360 Peripherals/Xbox 360 Controller for Windows

The Quanser Mechatronic Sensors Trainer conforms to the specifications for USB audio devices, a USB camera and a USB serial port so it may be used with operating systems and software that supports standard USB devices when it comes to those devices (microphones, speaker, camera and serial port). For example, the Camera application that comes with Windows supports Quanser Mechatronic Sensors Trainer out of the box, while PuTTy supports the serial port on Quanser Mechatronic Sensors Trainer out of the box. Likewise, the "Setup USB game controllers" Properties dialog supports Quanser Mechatronic Sensors Trainer as a game controller out of the box.

viii. SD Card Considerations

To use an SD Card with the Mechatronic Sensors Trainer, the SD card must be formatted with the FAT file system (FAT32) as opposed to exFAT. 32 GB or less tends to be FAT filesystem by default, anything larger is exFAT by default. However, we recommend formatting the SD card to the desired file system to be sure. To read an SD card larger than 32 GB that has been formatted as FAT, an updated Windows 11 computer is needed.

We recommend using a 32 GB SD card or smaller for this device. You can find information on how to format an SD card using a Windows computer here.

ix. Encoder Standards

Encoder outputs are assigned to match the standard across Quanser devices, which could lead to a mismatch with the datasheet. The standard in Quanser devices is B leads A for a CW direction.

When motors are used in the device, a positive motor input leads to a counterclockwise rotation of the shaft which increases the encoder counts. This matches all previously released Quanser legacy devices.

x. Camera Resolutions

The Quanser Mechatronic Sensors Trainer provides a color camera (OV5640) with up to 2592x1944 pixel resolution. The available camera resolutions are listed below:

Resolution	Description
160x120	Smallest resolution, 0.02MP 4:3 aspect ratio.
176x144	0.03MP 11:9 aspect ratio.
320x240	0.08MP 4:3 aspect ratio.
640x480	0.3MP 4:3 aspect ratio.
720x480	0.3MP 3:2 aspect ratio.
720x576	0.4MP 5:4 aspect ratio.
1024x768	0.8MP 4:3 aspect ratio.
1280x720	0.9MP 16:9 aspect ratio.
1920x1080	2.1MP 16:9 aspect ratio.
2592x1944	Maximum resolution of 5.0MP 4:3 aspect ratio.

C. Handling and Setup

i. Powering Up the Device

To set up the Mechatronic Sensors Trainer, follow these steps:

1. Make sure the Mechatronic Sensors Trainer is out of its case. Add the base underneath if needed, make sure the notch on the base matches the notch on the sensor side of the device.
2. Power the device up using the provided USB cable connected to the PC.
3. The device should start up with 3 blinks in blue and change colors quickly to a constant white.
4. The LED next to the USB C port will stay white after the display shows that it is loading.
5. The load screen with the Quanser Logo should disappear after a few seconds and you should see some evenly spaced crosses on a black background on the touch screen.

ii. Powering Up the Device Without a Computer

When the device is powered by a standalone power source, such as a USB power adapter or battery pack, and not connected to a computer (for example, a Windows PC or Raspberry Pi), the power-up procedure is the same as described in the previous section.

In this configuration where there is no computer detected, the device automatically enters **standalone recording mode**, which allows data from all sensors to be recorded directly to an SD card. On the bottom-right corner of the screen, a **REC** button will be displayed. If no compatible SD card is detected, the **REC** button appears greyed out. If an SD card is inserted and formatted according to the SD Card Considerations section (B.vi), the **REC** button becomes active, and the text will be white. Press **REC** to begin recording all signals to the SD card. Once recording has started, the button label changes to **STOP**. Press **STOP** to end the recording.

You may create multiple recordings during a session. Each recording is saved with a unique filename to prevent existing data from being overwritten.

iii. Powering Off the Device

The Mechatronic Sensors Trainer does not have a power off button. To power off the device, simply disconnect the USB cable the device is connected to.

iv. Low Power Mode

When using the device with the provided USB A to USB C cable, the screen will display a Low-Power mode warning on the display screen. To avoid this, use the provided USB C to USB C cable instead.

In low power mode, the maximum current that may be drawn from the user 3.3V and encoder 5V is 0.1 A total. In normal mode, it is 0.5 A. This only affects external sensors, all the onboard sensors are not affected when this warning is shown on the screen.

If the current limit is exceeded, you will get a fatal error message on the LCD requesting that you power down the device and resolve the issue.

D. Environmental

The Mechatronic Sensors Trainer is designed to function under the following environmental conditions:

- Standard rating
- Indoor use only
- Atmospheric conditions
    - Temperature - 15°C to 35°C
    - Altitude - up to 5000 m
    - Relative humidity - 30% to 60%
    - Air Pressure – 30 kPa (300 mbar) – 110 kPa (1100 mbar)
- Pollution Degree 2
- Mains supply voltage fluctuations up to 10% of nominal voltage
- Maximum transient overvoltage 2500 V

E. Electrical Considerations

	ESD warning	The Mechatronic Sensors Trainer is sensitive to electrostatic discharge. Before handling the device, ensure that you have been properly grounded.

	Caution	Use only the supplied USB-C cable to power the Mechatronic Sensors Trainer. When connected to a USB-A port, the available user current is limited to 100 mA. When connected to a USB-C port, the available user current is limited to 500 mA. Drawing more than the specified current may damage the host device and/or the trainer.

	Caution	Do not allow conductive materials to touch the sensor surfaces or the user I/O header pins. Contact with conductive objects can cause short circuits and permanent damage to the electronics of the device.

	Caution	The Mechatronic Sensors Trainer is not waterproof. Keep the device away from liquids and do not operate it in environments with high humidity, condensation, or risk of spills.

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